

The primal problem is

$$\min_{x \in \mathbb{R}^2} 4x_1 + 2x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &\geq 2, \\ -x_1 - x_2 &\geq 1, \\ x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

The quantity $x_1 + x_2$ must be both larger or equal to 2, and lesser or equal to -1. This is not possible, and the problem is infeasible. Consequently, we expect the dual problem to be either infeasible or unbounded.

To write the Lagrangian, we first write the problem as follows:

$$\min_{x \in \mathbb{R}^2} 4x_1 + 2x_2$$

subject to

$$\begin{aligned} g_1(x) &= 2 - x_1 - x_2 \leq 0 \quad (\mu_1), \\ g_2(x) &= 1 + x_1 + x_2 \leq 0 \quad (\mu_2), \\ g_3(x) &= -x_1 \leq 0 \quad (\mu_3), \\ g_4(x) &= -x_2 \leq 0 \quad (\mu_4), \end{aligned}$$

where μ_1, μ_2, μ_3 and μ_4 are the penalty parameters associated with each constraint.

The Lagrangian function is

$$\begin{aligned} L(x_1, x_2, \mu_1, \mu_2, \mu_3, \mu_4) &= 4x_1 + 2x_2 + \mu_1(2 - x_1 - x_2) \\ &\quad + \mu_2(1 + x_1 + x_2) - \mu_3x_1 - \mu_4x_2 \\ &= (4 - \mu_1 + \mu_2 - \mu_3)x_1 \\ &\quad + (2 - \mu_1 + \mu_2 - \mu_4)x_2 + 2\mu_1 + \mu_2. \end{aligned}$$

The dual function is

$$q(\mu_1, \mu_2, \mu_3, \mu_4) = \min_{x_1, x_2} L(x_1, x_2, \mu_1, \mu_2, \mu_3, \mu_4).$$

In order for the dual function to be bounded, the coefficients of x_1 and x_2 have to be zero. Therefore, we have to impose the following constraints on the penalty parameters:

$$\mu_3 = 4 - \mu_1 + \mu_2, \tag{1}$$

$$\mu_4 = 2 - \mu_1 + \mu_2. \tag{2}$$

In that case, the dual function becomes

$$q(\mu_1, \mu_2, \mu_3, \mu_4) = 2\mu_1 + \mu_2.$$

Moreover, the parameters associated with the inequality constraints must be non negative. That is, $\mu_1 \geq 0$, $\mu_2 \geq 0$, $\mu_3 \geq 0$, $\mu_4 \geq 0$. The dual problem is therefore

$$\max_{\mu_1, \mu_2, \mu_3, \mu_4} 2\mu_1 + \mu_2$$

subject to

$$\begin{aligned}\mu_3 &= 4 - \mu_1 + \mu_2, \\ \mu_4 &= 2 - \mu_1 + \mu_2, \\ \mu_1 &\geq 0, \\ \mu_2 &\geq 0, \\ \mu_3 &\geq 0, \\ \mu_4 &\geq 0.\end{aligned}$$

As the parameters μ_3 and μ_4 do not appear in the objective function, we can rewrite (1) and (2) as

$$\begin{aligned}4 - \mu_1 + \mu_2 &\geq 0, \\ 2 - \mu_1 + \mu_2 &\geq 0,\end{aligned}$$

and eliminate μ_3 and μ_4 , so that the dual problem becomes

$$\max_{\mu_1, \mu_2} 2\mu_1 + \mu_2$$

subject to

$$\begin{aligned}4 - \mu_1 + \mu_2 &\geq 0, \\ 2 - \mu_1 + \mu_2 &\geq 0, \\ \mu_1 &\geq 0, \\ \mu_2 &\geq 0.\end{aligned}$$

The problem happens to be unbounded. To see it, first fix the value of μ_1 to 0. We obtain

$$\max_{\mu_2} \mu_2$$

subject to

$$\begin{aligned}\mu_2 &\geq -4, \\ \mu_2 &\geq -2, \\ \mu_2 &\geq 0.\end{aligned}$$

As μ_2 is not constrained from above, its value and, therefore, the value of the objective function, can always be made larger than any arbitrary value.